

Network Science HS23

Assignment 3

Blockchain & Distributed Ledger Technologies Group

FOR STUDIES PURPOSES ONLY

Instructions:

1. You must submit your answers through OLAT.
2. You have three weeks to submit the results. The deadline is 04.12.2023.
3. Answer means
 - Code or theoretical solution, depending on the exercise request.
 - A document with the results clearly visible.
 - Possible formats are PDF, Python Notebooks.
 - All code must be already executed and results clearly visible.
4. Show all your work and reasoning clearly.
5. You can do this assignment in groups of up to two.

1 ERGMs

For each of the World Trade Web (WTW) networks provided:

1. Compute the strength assortativity coefficient on the data.

Hint: Pay attention: the underlying data (WTW) are weighted and directed, meaning you should account for four different kind of assortativity: in-in, in-out, out-in, out-out.

2. Fit the Undirected Enhanced CM and Directed Enhanced CM using the CReMa method.

Hint: You can use the nemtropy package on networkx to deploy these methods. You can find some examples at <https://github.com/nicoloval/NEMtropy/blob/master/examples/Directed%20Graphs.ipynb>. For additional questions, do not hesitate to ask at the QA.

3. Sample 30 networks from the obtained ERGM distributions and measure the strength assortativity on all of them. Calculate average and standard deviation of each assortativity typology.

4. Plot strength assortativity as a function of time (the WTW network year), comparing the observed real value with the average and error bars obtained from the samples.

Hint: plot error bars as confidence intervals. Plot all pairs of assortativity for the undirected and directed case(in-in, in-out, out-out)

5. Write a short paragraph to comment on the interpretation of strength assortativity for this dataset, and about the conclusions you can draw via the inference you made using the ERGM models.

Datasets provided:

- WDN_XXXX.txt.graphml: Data is in graphml format. Networks are directed and edges (i, j) have a weight attribute which represents the value of trade from i to j in million US\$, data is aggregated yearly, ranging from 1992 to 2002 Gleditsch [2002].

2 Stock correlations

Answer the following questions using the dataset provided (described at the end):

1. Is there a market mode at the minute and hour timescale?

Hint: The existence of the market mode is related to the gap between the largest and second largest eigenvalues of the correlation matrix.

2. Compute the degree distributions of the 1 hour and 1 minute MSTs. Which are the 5 stocks with highest degree on the two different timescales MST?

Hint: To build the MST, use `networkx.minimum_spanning_tree` on the weighted undirected graph that is generated by the matrix with entries d_{ij} (see lecture on financial networks).

Assign the correct ticker (AAPL, AMZN, ...) as an attribute to the nodes, by using

```
networkx.set_node_attributes(G, values=tickers, name='ticker')
```

3. Plot the degree distributions at 1 minute and 1 hour. Are the degree distributions qualitatively the same? Provide a comment to explain eventual similarities or dissimilarities.
4. Compare the MST degree distributions of the 1 hour, 1 minute, Normal and One-Factor correlations: for each time scale, plot the three different distributions on the same plot. Is the degree distribution of MSTs well reproduced by the null Gaussian and One-Factor models?

Datasets provided:

- `cormat_1h.npy` and `cormat_1m.npy` are Correlation matrices corresponding to the 1-hour and 1-minute correlations of log-returns in open market hours, for the 97 most capitalized stocks on the New York Stock Exchange in September 2014.
- `cormat_gaussian_1m.npy` and `cormat_gaussian_1h.npy`, are correlation matrices for the corresponding Gaussian models opportunely fitted to the data.
- `cormat_onefactor_1m.npy` and `cormat_onefactor_1h.npy`, are correlation matrices for the corresponding One-Factor models opportunely fitted to the data.

Source: NYSE Consolidated Trades dataset, accessed through UZHs Wharton Research Data Services platform. Rows and columns are sorted according to the `stocknames.txt` file list, containing stock tickers (trading codes).

3 Robustness

1. Generate synthetic networks (you can use whatever module you prefer):
 - Random networks (ErdosRényi) (with $n = 1000$ and $\langle k \rangle = \{1, 4\}$)
 - Scale-free networks (Barabási-Albert) (with $n=1000$ and $m = \{2, 4\}$)
2. Plot the degree distributions of the synthetic networks and of the networks provided in the datasets.
3. Plot the size of the largest connected component as a function of removed edges and nodes in two scenarios for the synthetic graphs and the real networks in the datasets:
 - (a) Random attacks (failures)
 - (b) Targeted attacks (degree centrality and betweenness centrality)
4. Comment the plots, using what we discussed in class.

Datasets provided:

- `graph_eu_airlines.gml`: European airline network. Nodes represent airlines and Edges represent airline routes among European airports Cardillo et al. [2013]
- `graph_power.gml`: The file contains an undirected unweighted representation of the topology of the Western States Power Grid of the United States Watts and Strogatz [1998]

References

- Cardillo, A., Gómez-Gardenes, J., Zanin, M., Romance, M., Papo, D., del Pozo, F. and Boccaletti, S. [2013]. Emergence of network features from multiplexity, *Scientific Reports* **3**: Article number: 1344.
- Gleditsch, K. S. [2002]. Expanded trade and gdp data, *Journal of Conflict Resolution* **46**(5): 712–724.
- Watts, D. J. and Strogatz, S. H. [1998]. Collective dynamics of small-world networks, *nature* **393**(6684): 440–442.